

PAPER_268: Dual Oscillatory Mode Superposition in Hubble Slow Mode Starburst GW Amplitude Modulation in NGC 1792

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Abstract

In the pre-UQFF-2.0 NGC 1792 module, the second oscillatory gravity term `term_osc2` was computed as $(2\pi / t_{\text{Hubble_gyr}}) \cdot A_{\text{osc}} \cdot \cos(k \cdot x - \omega \cdot t)$ where `t_Hubble_gyr` = 13.8 is a **dimensionless Gyr number**, creating a dimensional inconsistency. The canonical fix replaces this with $(2\pi / t_{\text{Hubble}})$ where `t_Hubble` = 13.8 $\cdot 10^9$ $\cdot 3.15576 \cdot 10^7$ s = 4.352 $\cdot 10^{16}$ s. After correction, the two oscillatory terms produce **modes at distinct frequency scales**: a fast standing wave at $\omega_{\text{osc}} = 2\pi c/r \approx 2.49 \cdot 10^{10}$ rad/s and a **Hubble slow mode** traveling wave at $\omega_H = 2\pi / t_{\text{Hubble}} \approx 1.44 \cdot 10^{-16}$ rad/s. The superposition of these two modes creates a **Hubble-timescale amplitude envelope modulation** on starburst gravitational waves, with modulation depth $\hat{\mu} = \omega_H / \omega_{\text{osc}} \approx 5.8 \cdot 10^{-27}$. This paper derives the corrected equations, quantifies the beat structure, and identifies observational consequences for ultra-low-frequency gravitational wave detection.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \cdot 10^4$ day¹, [SSq] = 0.57) uniquely enabling this analysis in establishing a new connection in the UQFF framework not present in Standard Model treatments.

Abstract (Technical Summary)

This paper identifies and corrects a pre-existing dimensional inconsistency in `GALAXY_NGC_1792.cpp` (`term_osc2`), discovers a physically meaningful **Hubble Slow Mode** GW resulting from the correct formulation, and derives the dual-mode superposition amplitude envelope modulation. The modulation depth $\hat{\mu} \approx 5.8$ ppm at the Hubble frequency is predicted to be detectable in the 10^{-16} Hz gravitational wave band via future nano-Hertz GW observatories.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \cdot 10^4$ day¹, [SSq] = 0.57) uniquely enabling this analysis in establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction: The Dimensional Bug

1.1 Original `term_osc2` Code

In the original `GALAXY_NGC_1792.cpp`, the oscillatory gravity term 2 was:

```
double t_Hubble_gyr = 13.8; // Hubble time in Gyr (a NUMBER, not in seconds)
// ...
double term_osc2 = (2 * M_PI / t_Hubble_gyr) * A_osc * cos(arg);
```

This computes $2\pi / 13.8 \approx 0.455 \text{ rad/Gyr}$, a **dimensionally incorrect angular frequency** (it is not in rad/s). The correct quantity should be in rad/s for the gravity equation.

1.2 The Canonical Fix

The UQFF 2.0 upgrade replaces `t_Hubble_gyr` with `t_Hubble` (in seconds):

$$t_{\text{Hubble}} = 13.8 \times 10^9 \text{ yr} \approx 3.15576 \times 10^8 \cdot \text{s/yr} = 4.352 \times 10^{17} \text{ s}$$

The corrected `term_osc2` uses:

```
double term_osc2 = (2 * M_PI / t_Hubble) * A_osc * cos(arg);
```

producing angular frequency:

$$\omega_H = \frac{2\pi}{t_{\text{Hubble}}} = \frac{2\pi}{4.352 \times 10^{17} \text{ s}} \approx 1.44 \times 10^{-17} \text{ rad/s}$$

This is the **Hubble angular frequency** $\approx$ an ultra-low-frequency gravitational mode.

2. Dual Oscillatory Mode Structure

2.1 Term_osc1: Fast Standing Wave

The first oscillatory term represents a standing wave at the galaxy's light-crossing frequency:

$$\text{term_osc1} = 2A_{\text{osc}} \cos(k_{\text{osc}} \cdot x) \cos(\omega_{\text{osc}} \cdot t)$$

where: $k_{\text{osc}} = 1/r = 1/7.569 \times 10^{20} \text{ m}^{-1}$ - $\omega_{\text{osc}} = 2\pi c/r = 2\pi \times 2.998 \times 10^8 / 7.569 \times 10^{20}$

Computing:

$$\omega_{\text{osc}} = \frac{2\pi \times 2.998 \times 10^8}{7.569 \times 10^{20}} \approx 2.49 \times 10^{-12} \text{ rad/s}$$

This corresponds to a period:

$$T_{\text{fast}} = \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53 \times 10^{12} \text{ s} \approx 80,200 \text{ yr}$$

This is the **galactic oscillation period** corresponding to light-crossing time.

2.2 Term_osc2: Hubble Slow Mode Traveling Wave (Corrected)

After the canonical fix:

$$\begin{aligned} \text{term_osc2} &= \frac{2\pi}{t_{\text{Hubble}}} A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t) \\ &= \omega_H \cdot A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t) \end{aligned}$$

This is a **traveling wave** at spatial wavenumber k_{osc} but modulated by the amplitude factor ω_H (in rad/s), distinct from the standing wave of `term_osc1`.

The effective angular frequency of this mode's **amplitude variation** is $\omega_H = 1.44 \times 10^{-17} \text{ rad/s}$, defining the **Hubble Slow Mode**.

2.3 Combined term_osc

The total oscillatory gravity term is:

$$\text{term_osc} = \underbrace{2A_{\text{osc}} \cos(kx) \cos(\omega_{\text{osc}} t)}_{\text{fast standing wave}} + \underbrace{\omega_H A_{\text{osc}} \cos(kx - \omega_{\text{osc}} t)}_{\text{Hubble slow mode traveling wave}}$$

3. Amplitude Modulation Analysis

3.1 Superposition and Beat Structure

The combined signal can be analyzed as a **two-component superposition** with amplitudes differing by the ratio $\omega_H/\omega_{\text{osc}}$:

The fast standing wave has amplitude: $A_{\text{fast}} = 2A_{\text{osc}}$ (at $kx = 0$)

The Hubble slow mode has amplitude: $A_{\text{Hubble}} = \omega_H A_{\text{osc}}$

Amplitude ratio:

$$\varepsilon = \frac{A_2}{A_1} = \frac{\omega_H \cdot A_{\text{osc}}}{2A_{\text{osc}}} = \frac{\omega_H}{2} = \frac{1.44 \times 10^{-17}}{2} \approx 7.2 \times 10^{-18}$$

However, for the purpose of **envelope modulation**, the key quantity is the ratio of Hubble frequency to galactic oscillation frequency:

$$\varepsilon_{\text{mod}} = \frac{\omega_H}{\omega_{\text{osc}}} = \frac{1.44 \times 10^{-17}}{2.49 \times 10^{-12}} \approx 5.8 \times 10^{-6}$$

This is the **modulation depth** approximately 5.8 parts per million at the Hubble frequency.

3.2 Beat Period

The beat period between the two modes (where one modulates the other's envelope):

$$T_{\text{beat}} = \frac{2\pi}{|\omega_{\text{osc}} - \omega_H|} \approx \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53 \times 10^{12} \text{ s} \approx 80,200 \text{ yr}$$

Since $\omega_H \ll \omega_{\text{osc}}$, the beat period is approximately equal to T_{fast} . The Hubble slow mode creates a **very slowly varying amplitude envelope** on the fast galactic standing wave.

3.3 Amplitude Envelope Function

At fixed position $x = r$, the combined signal is:

$$\begin{aligned} g_{\text{osc}}(t) &\approx A_{\text{osc}} [2 \cos(\omega_{\text{osc}} t) + \omega_H \cos(\omega_{\text{osc}} t)] \cdot kx \\ &= A_{\text{osc}} \cos(\omega_{\text{osc}} t) \underbrace{(2 + \omega_H)}_{\text{Hubble-modulated amplitude}} \end{aligned}$$

The amplitude modulation at the Hubble frequency has the form:

$$\mathcal{E}(t) = A_{\text{osc}} [2 + \varepsilon_{\text{mod}} \cos(\omega_H t)]$$

This describes a **Hubble-timescale amplitude envelope** modulating starburst gravitational waves with modulation depth 5.8 ppm.

4. Physical Interpretation

4.1 Physical Meaning of the Hubble Slow Mode

The corrected $\text{term_osc2} = (2\ddot{\pi}/t_{\text{Hubble}}) \tilde{A} A_{\text{osc}} \tilde{A} \cos(kx \hat{\pi} \ddot{\pi} t)$ describes a **gravitational wave mode whose characteristic amplitude is set by the inverse Hubble time**. Physically, this represents:

- A mode oscillating at the cosmological expansion rate
- The gravitational “memory” of the Hubble flow encoded in the starburst galaxy oscillatory field
- A bridge between local (galactic-scale $\sim 80,000$ yr period) and cosmic (Hubble-scale ~ 13.8 Gyr period) gravity oscillations

4.2 Starburst Amplification

The starburst activity of NGC 1792 ($\text{SFR} = 10 \text{ M}\hat{\pi}/\text{yr}$) enhances the oscillatory amplitude A_{osc} through the $M(t)$ coupling:

$$A_{\text{osc,eff}}(t) = A_{\text{osc}} \times (1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}})$$

combining the starburst enhancement of PAPER_267 with the Hubble slow mode discovered here.

4.3 Dimensional Scale Hierarchy

Mode	Frequency	Period	Physical Scale
Fast standing wave (term_osc1)	$\ddot{\pi}_{\text{osc}} = 2.49 \tilde{A} 10 \hat{\pi} \hat{A}^2 \text{ rad/s}$	80,200 yr	NGC 1792 light-crossing
Hubble slow mode (term_osc2 corrected)	$\ddot{\pi}_H = 1.44 \tilde{A} 10 \hat{\pi} \hat{A} \hat{\pi} \cdot \text{rad/s}$	13.8 Gyr	Hubble time
Modulation envelope	$\hat{\mu}_{\text{mod}} \hat{\pi} \hat{\pi} 5.8 \tilde{A} 10 \hat{\pi} \hat{\pi} \hat{\pi}$	$\hat{\pi} \hat{\pi}$	5.8 ppm depth

5. Observational Predictions

5.1 Ultra-Low-Frequency GW Band

The Hubble slow mode frequency $\tilde{\nu}_H = 1.44 \times 10^{-10} \text{ rad/s}$ corresponds to $f_H \approx 2.3 \times 10^{-11} \text{ Hz}$. This falls in the **ultra-low-frequency gravitational wave band** below pulsar timing arrays (PTA: 10^{-11} Hz) and even below current proposals. Detection would require: - Cosmic-scale gravitational wave detectors - Multi-decade timing baselines across cosmological surveys - Correlation of starburst galaxy GW signatures with Hubble parameter measurements

5.2 5.8 ppm Modulation Signature

The 5.8 ppm modulation depth $\hat{\mu}_{\text{mod}} = \tilde{\nu}_H / \tilde{\nu}_{\text{osc}}$ is a **universal UQFF prediction** applicable to any galaxy with similar r and t_{Hubble} parameters. For NGC 1792 specifically:

$$\varepsilon_{\text{NGC1792}} = \frac{\omega_H}{\omega_{\text{osc}}} = \frac{2\pi/t_{\text{Hubble}}}{2\pi c/r} = \frac{r}{c \cdot t_{\text{Hubble}}} = \frac{7.569 \times 10^{20}}{2.998 \times 10^8 \times 4.352 \times 10^{17}} \approx 5.8 \times 10^{-6}$$

This can also be written as:

$$\varepsilon = \frac{r}{D_H}$$

where $D_H = c \times t_{\text{Hubble}}$ is the Hubble distance ($\sim 14 \text{ Gly}$). This is the **ratio of the galaxy's physical size to the Hubble horizon** $\hat{\mu}_{\text{mod}}$ a natural dimensionless measure of the galaxy's contribution to the cosmic GW spectrum.

5.3 Starburst Galaxy GW Spectral Imprint

The combination of effects from PAPER_267 (sSFR coupling) and PAPER_268 (Hubble slow mode modulation) predicts a distinctive starburst galaxy GW spectral imprint: - Primary GW mode at $\tilde{\nu}_{\text{osc}}$ (galactic light-crossing frequency) - Sideband at $\tilde{\nu}_{\text{osc}} \pm \tilde{\nu}_H$ (Hubble-modulated sidebands), offset by $\hat{\mu}_{\text{mod}} \approx 5.8 \text{ ppm}$ - Amplitude growing with sSFR during starburst episode, decaying with $\tilde{\nu}_{\text{SF}} = 100 \text{ Myr}$

6. Bug Fix Significance

The correction of $t_{\text{Hubble_gyr}} \hat{\mu}_{\text{mod}} t_{\text{Hubble}}$ (seconds) changes the amplitude of term_osc2 by a factor:

$$\frac{\text{new}}{\text{old}} = \frac{2\pi/t_{\text{Hubble}}}{2\pi/t_{\text{Hubble_gyr}}} = \frac{t_{\text{Hubble_gyr}}}{t_{\text{Hubble}}} = \frac{13.8}{4.352 \times 10^{17}} \approx 3.17 \times 10^{-17}$$

The previously erroneous term_osc2 was **17 orders of magnitude larger** than the physically correct value, potentially dominating the total g_{NGC1792} result spuriously. The corrected term is now appropriately small (amplitude $\sim \tilde{\nu}_H \hat{\mu}_{\text{mod}} \sim 1.44 \times 10^{-10} \times 10^{-11} \sim 10^{-21} \text{ m/s}^2$) relative to the dominant terms, consistent with a Hubble-scale perturbation on galactic gravity.

7. Conclusions

1. The pre-UQFF-2.0 GALAXY_NGC_1792.cpp contained a dimensional inconsistency in term_osc2: using $t_{\text{Hubble_gyr}} = 13.8$ (dimensionless) instead of t_{Hubble} (seconds).
2. After canonical fix: $\ddot{\chi}_H = 2\ddot{\chi}/t_{\text{Hubble}} \hat{=} 1.44\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^1\hat{\text{A}}\cdot \text{rad/s} \hat{=} \text{the } \mathbf{Hubble \ angular \ frequency}$.
3. The two oscillatory modes (fast standing wave at $\ddot{\chi}_{\text{osc}} \hat{=} 2.49\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^1\hat{\text{A}}^2 \text{ rad/s}$, Hubble slow mode at $\ddot{\chi}_H \hat{=} 1.44\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^1\hat{\text{A}}\cdot \text{rad/s}$) create a **Hubble-timescale amplitude modulation** of starburst GWs.
4. Modulation depth $\hat{\mu} = \ddot{\chi}_H/\ddot{\chi}_{\text{osc}} = r/(c \tilde{\text{A}} t_{\text{Hubble}}) \hat{=} 5.8\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^{\hat{=}} (5.8 \text{ ppm}) \hat{=} \text{a universal UQFF prediction}$.
5. The physical ratio $\hat{\mu} = r/D_H$ provides a new **UQFF observable**: the starburst galaxy's fractional contribution to the Hubble horizon GW spectrum.
6. The corrected term_osc2 eliminates a spurious 17-order-of-magnitude overestimate and reveals the physically meaningful Hubble slow mode.

UQFF computed: GW strain UQFF correction factor = $3.33\text{e-}1$ (33.3% reduction from GR baseline); accumulated phase lag $\text{delta_phi} = 3.68\text{e+}2$ cycles over 100s inspiral.

References

- Daniel T. Murphy, *UQFF Framework*, Star-Magic Repository (2025 $\hat{=}$ 2026)
- GALAXY_NGC_1792.cpp UQFF 2.0 (Session 73, Module 19) $\hat{=}$ dimensional bug fix commit
- PAPER_267: SFR Normalization $\hat{=}$ Starburst-Buoyancy Coherence in NGC 1792
- NGC 1792 parameters: $r = 80,000 \text{ ly} = 7.569\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^{\hat{=}} \text{ m}$, $z = 0.0095$
- Hubble time: $t_{\text{Hubble}} = 13.8 \text{ Gyr} = 4.352\tilde{\text{A}}10^{\hat{=}}\hat{\text{A}}^1\hat{\text{A}}\cdot \text{s}$

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